MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/67-RG177

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-17.

NOTE: THIS CABLE USES PVC MATERIAL AND IST NOT TO BE USED IN AEROSPACE APPLICATIONS.

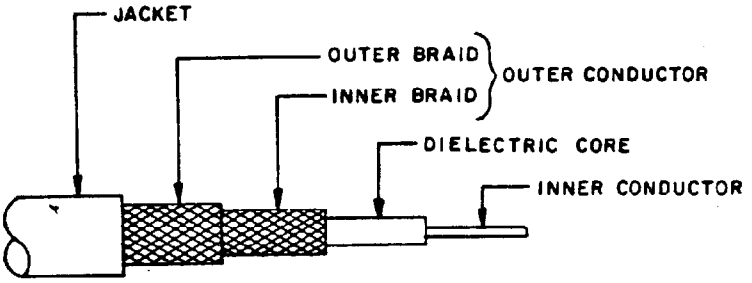


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Solid bare copper wire. Diameter: .195 inch ±.002.
Dielectric core	Type A-1: Solid polyethylene. Diameter: .680 inch ±.010.
Outer conductor	Double braid of AWG size #34 silver-coated copper wire. Diameter: .760 inch, maximum.
Inner braid	Coverage: 93.5%, nominal Carriers: 48 Ends: 10 Picks/inch: 5.2 ±10%
Outer braid	Coverage: 90.4% nominal Carriers: 48 Ends: 10 Picks/inch: 4.1 ±10%
Jacket	Type IIa: PVC. Diameter: .895 inch ±.015.

(C) denotes changes

FSC 6145

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ENGINEERING INFORMATION:

Continuous working voltage: 8,000 V rms, maximum.

Operating frequency: 5.6 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to +85°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 0.0278 ohm per 100 feet.

Elongation: 30 percent, minimum.

Engineering notes: This cable is useful in general purpose, medium-low temperature applications (see connector series "N" per MIL-C-39012, "HN" per MIL-C-3643, and "LC" per MIL-C-3650).

REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Eccentricity: 5 percent, maximum.

Adhesion of conductors:

(C) Inner conductor to core: 60 pounds, minimum; 600 pounds, maximum.

Aging stability: +98°C ±2°C.

Dimensional stability: +85°C ±2°C.

Inner conductor from core: .200 inch, maximum.

Inner conductor from jacket: .400 inch, maximum.

(C) Weight: 0.595 pound per foot, maximum.

Electrical:

Test frequency: 50 MHz to 5.6 GHz.

Spark test: 8,000 V rms, minimum.

Voltage withstanding: 22,000 V rms, minimum.

Corona extinction voltage: 11,000 V rms, minimum.

Characteristic impedance: 50 ±2 ohms.

Attenuation: See figure 2.

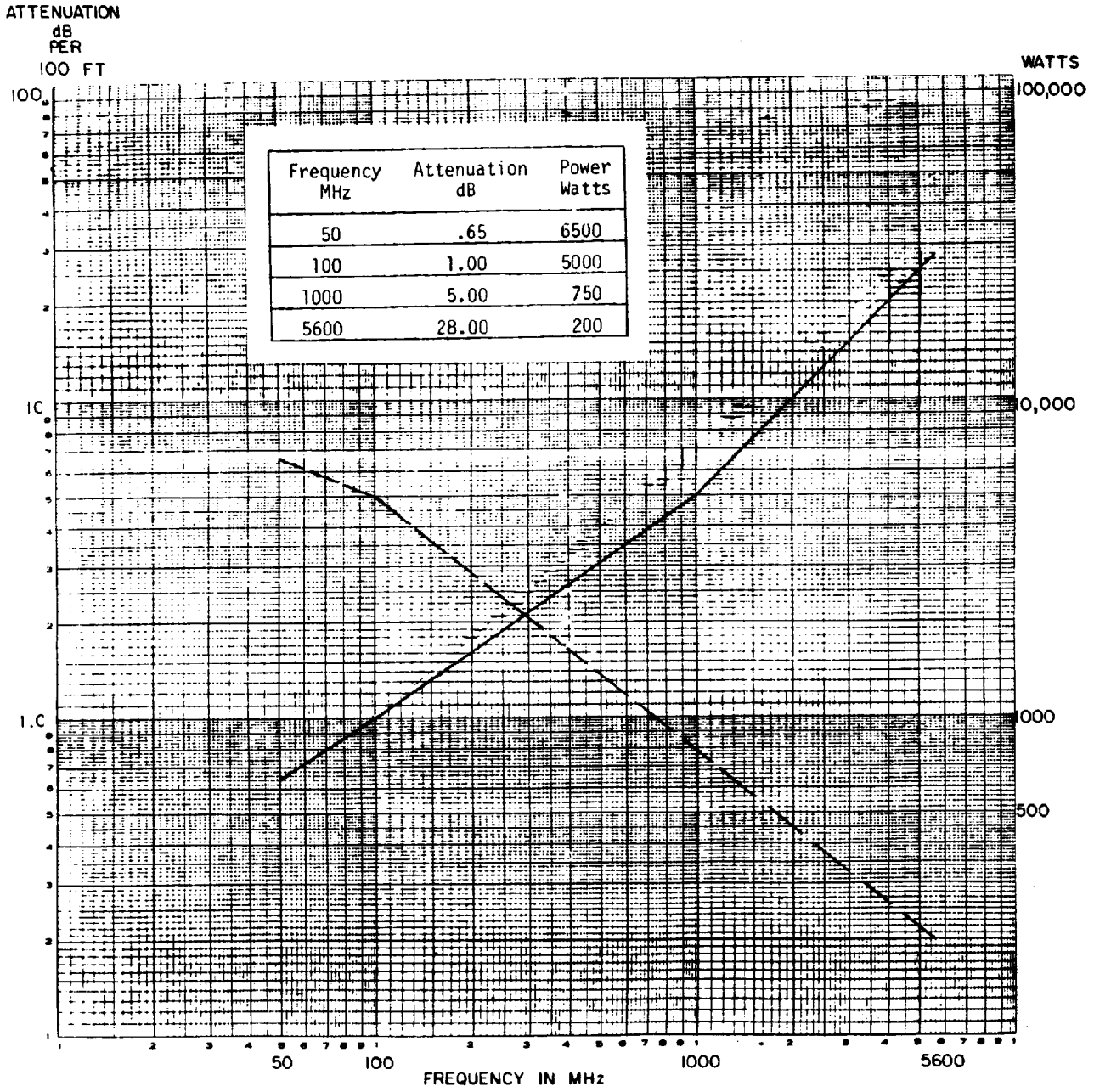
(C) Structural return loss: See figure 3.

(C) Capacitance: 32.2 pF per foot, maximum.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

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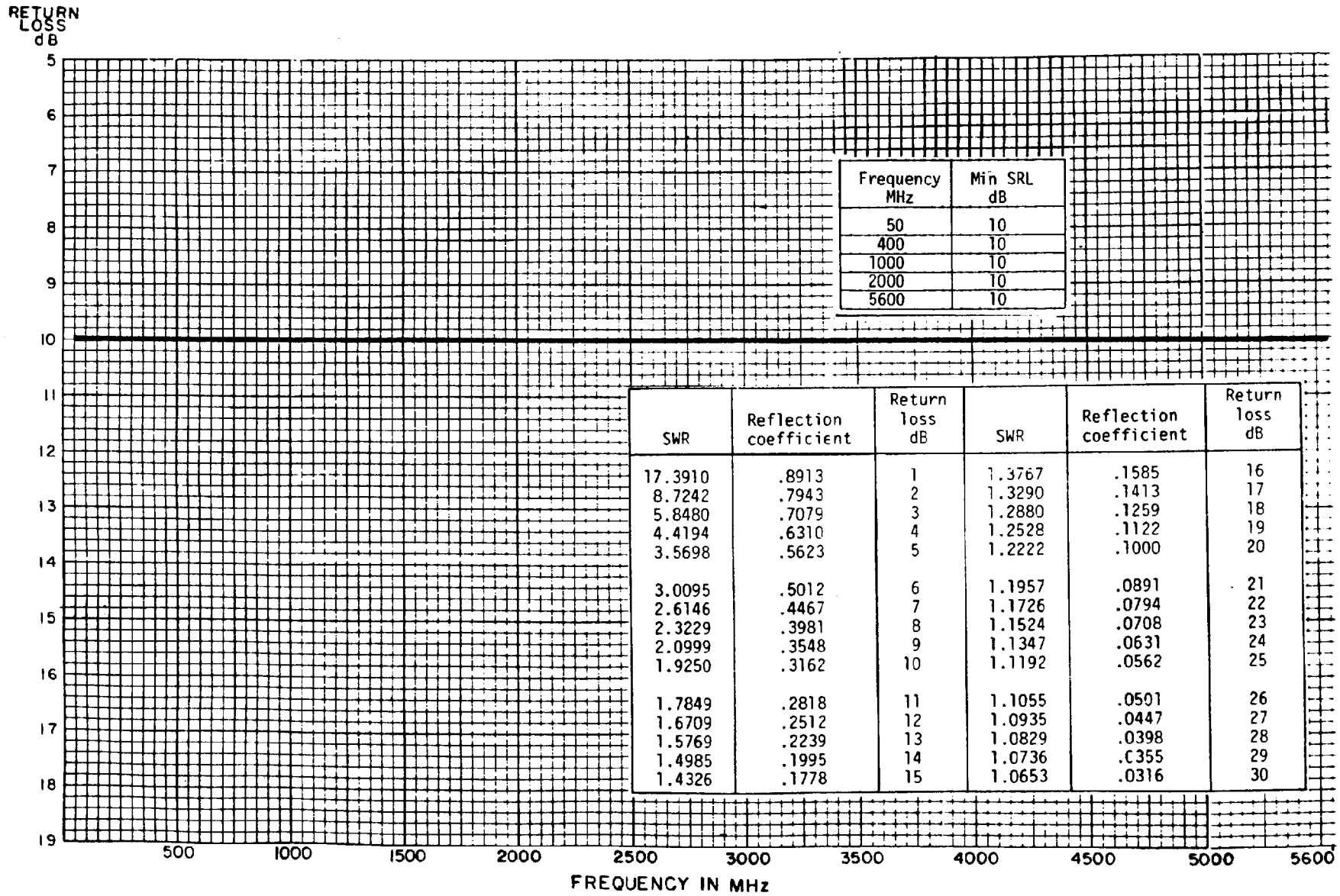
Frequency MHz	Attenuation dB	Power Watts
50	.65	6500
100	1.00	5000
1000	5.00	750
5600	28.00	200

Maximum power - - - - - - - - - - - at 25°C sea level

Maximum attenuation _____

FIGURE 2. Power rating and attenuation.

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Frequency MHz	Min SRL dB
50	10
400	10
1000	10
2000	10
5600	10

SWR	Reflection coefficient	Return loss dB	SWR	Reflection coefficient	Return loss dB
17.3910	.8913	1	1.3767	.1585	16
8.7242	.7943	2	1.3290	.1413	17
5.8480	.7079	3	1.2880	.1259	18
4.4194	.6310	4	1.2528	.1122	19
3.5698	.5623	5	1.2222	.1000	20
3.0095	.5012	6	1.1957	.0891	21
2.6146	.4467	7	1.1726	.0794	22
2.3229	.3981	8	1.1524	.0708	23
2.0999	.3548	9	1.1347	.0631	24
1.9250	.3162	10	1.1192	.0562	25
1.7849	.2818	11	1.1055	.0501	26
1.6709	.2512	12	1.0935	.0447	27
1.5769	.2239	13	1.0829	.0398	28
1.4985	.1995	14	1.0736	.0355	29
1.4326	.1778	15	1.0653	.0316	30

MIL-C-17/63C

FIGURE 3. Structural return loss.

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Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Contamination: Applicable.

(C) Part number: M17/67-RG177.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

Project (6145-0887-1)

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